Show all steps for each conversion. Clearly identify the number system of each answer using the appropriate base (subscript).

1. Convert each of the following binary numbers to its octal, decimal, and hexadecimal equivalents.

111011101_2

10101010111_2

111100000_2

2. Convert each of the following octal numbers to its binary, decimal, and hexadecimal equivalents.

3754_8

7777_8

247_8

3. Convert each of the following decimal numbers to its binary, octal, and hexadecimal equivalents.

3479_{10}

642_{10}

555_{10}

4. Convert each of the following hexadecimal numbers to its binary, octal, and decimal equivalents.

$4FB2_{16}$

$88BAE_{16}$

$DC4_{16}$

Answer Key

Original Number	Binary	Octal	Decimal	Hexadecimal
111011101_2	111011101	735	477	1DD
10101010111_2	10101010111	2527	1367	557
111100000_2	111100000	740	480	1E0
3754_8	11111101100	3754	2028	7EC
7777_8	111111111111	7777	4095	FFF
247_8	10100111	247	167	A7
3479_{10}	110110010111	6627	3479	D97
642_{10}	1010000010	1202	642	282
555_{10}	1000101011	1053	555	22B
$4FB2_{16}$	100111110110010	47662	20402	4FB2
$88BAE_{16}$	10001000101110101110	2105656	560046	88BAE
$DC4_{16}$	110111000100	6704	3524	DC4